

Clinical Signs, Pathophysiology and Management of Cutaneous Side Effects of Anti-Tumor Necrosis Factor Agents

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Abstract Approximately one in four patients treated with anti-TNF agents (infliximab, etanercept, adalimumab, certolizumab, and golimumab) develops cutaneous adverse events, typically months to years after the initiation of treatment, with xerosis cutis, eczema (often psoriasiform), psoriasis, palmoplantar pustulosis, cutaneous infections, alopecia, and skin cancer being the most frequently encountered. The typical skin lesion of anti-tumor necrosis factor (TNF)-treated patients is orange–red psoriasiform eczema affecting the flexures, genitalia, scalp, or face, with high susceptibility to bacterial superinfection with *Staphylococcus aureus*. When adequate dermatological treatment is administered to patients with skin lesions receiving anti-TNF treatment, the discontinuation of anti-TNF agents is only rarely required. Smoking, female sex, and Crohn’s disease are most frequently observed as risk factors for anti-TNF-induced cutaneous adverse events. The underlying pathophysiology is still poorly understood, but epidermal permeability barrier dysfunction, increased susceptibility to bacterial superinfection, and cytokines derived from T helper (Th) 1 (interferon- γ), Th17 cells (interleukin [IL]-17A and IL-22), plasmacytoid dendritic cells (interferon- α), and keratinocytes (IL-36 γ and IL-17C) appear to play a role. In this review, we describe the clinical characteristics, risk factors, pathophysiology, and management of cutaneous adverse events of patients

treated with anti-TNF agents. In addition, we try to give some practical guidance on how to prevent and manage the skin changes in anti-TNF-treated patients, based on our own experience with dermatological care in a large cohort of inflammatory bowel disease patients receiving anti-TNF therapy.

Key Points

A substantial number of patients receiving anti-tumor necrosis factor (TNF) treatment present with skin lesions, most often dry skin, (psoriasiform) eczema, psoriasis, palmoplantar pustulosis, skin infection, alopecia, or skin cancer.

Anti-TNF-induced (psoriasiform) eczema is often superinfected with *Staphylococcus aureus*, which can be treated with topical antibiotics/antiseptics and/or appropriate systemic antibiotics.

Discontinuation of anti-TNF therapy is only rarely required upon adequate dermatological treatment of anti-TNF-induced skin lesions.

1 Introduction

Anti-tumor necrosis factor (TNF) biologics have revolutionized the treatment of rheumatoid arthritis (RA), inflammatory bowel disease (IBD), and psoriasis. The soluble receptor etanercept is only indicated for arthritis and psoriasis, whereas the more powerful TNF antibodies (or antibody fragments) infliximab, adalimumab, golimumab, and certolizumab can also be used for IBD [1].

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Recently, adalimumab was also approved for the treatment of hidradenitis suppurativa [61].

As with other targeted therapies (e.g. epidermal growth factor receptor [EGFR] inhibitors used in oncology), interference with the physiological function of the target protein may result in mechanism-dependent, class-specific side effects [2]. As TNF plays a crucial physiological role in granuloma formation and maintenance, the flare of latent tuberculosis was rapidly identified as an on-target side effect of TNF inhibitors [3]. Gastroenterologists at our institution, who were already treating large numbers of Crohn's disease patients with infliximab in the early 2000s, noticed remarkable cutaneous side effects in a substantial number of their patients [4], and systematically started sending patients with skin problems to one dermatologist (SS). This collaborative effort of the Department of Gastroenterology and Dermatology at the University Hospitals in Leuven resulted in us gaining considerable insight into the cutaneous side effects of anti-TNFs and their management [5, 6]. The original data of the Leuven cohort relating to skin findings in IBD patients receiving anti-TNF therapy (including patient demographics, anti-TNF dosing, and statistics) were published 1 year ago [6].

In this literature review, we describe the possible dermatological problems in patients treated with anti-TNF agents. Furthermore, we propose a pragmatic treatment scheme, based on our own experience with dermatological care in a large cohort of IBD patients receiving anti-TNF therapy [6].

2 Clinical Findings

2.1 Prevalence

At our institution, 29% of a cohort of 917 IBD patients started on infliximab developed skin adverse events [6]. Most of these were new-onset skin lesions, and only a minority were an exacerbation of an existing skin disease [8]. In our cohort, the median time to develop skin lesions following the start of anti-TNF therapy was 1.9 years [6], which is in line with other studies reporting a median duration of anti-TNF therapy before the onset of cutaneous adverse events of 6 months [23], 7 months [8], 11 months [19, 21], 10–30 months [18], 13 months [20], 18.5 months [58], and 34–43 months [10]. However, cutaneous reactions such as eczema and psoriasis may occur as early as 1 week after the initiation of the anti-TNF agent [8]. Nevertheless, the usual long delay is a possible reason that the association with drug treatment is not always made, which may lead to underreporting of these skin signs. Similar rates of cutaneous adverse events of over 20% have been reported in other cohort studies with different TNF

inhibitors in arthritis and IBD (Table 1) [7–12]. Forty-eight percent of children with IBD receiving infliximab presented with skin lesions [13]. Two rather small studies (50 and 30 patients, respectively) even reported that not less than 60% of IBD patients developed skin reactions when taking adalimumab or infliximab [14, 15]. Psoriasis patients may also be affected by cutaneous side effects (especially palmoplantar pustulosis, guttate psoriasis, or psoriasiform dermatitis) [16, 17]. In psoriasis patients, these side effects are probably often misinterpreted as a flare of the existing psoriasis or lack of response to the TNF antagonist. Very recently, psoriasiform skin reactions were also described in 19.5% of hidradenitis suppuriva patients treated with anti-TNF agents [59].

A number of cohort studies in IBD patients looked at psoriasis only as an adverse event, with rates from 1.7 to 10.4% [18–22]. The British Society for Rheumatology Biologics Register reported new-onset psoriasis in 0.3% of anti-TNF-treated RA patients [23]. Finally, one study focusing on eczema only, recorded dermatitis in 16.3% of a prospective cohort of patients receiving infliximab for IBD or rheumatic disease [24].

2.2 Risk Factors

In the Leuven cohort of IBD patients initiated on infliximab, no relation was seen between skin adverse events and sex, type of IBD (Crohn's disease vs. ulcerative colitis), infliximab trough levels, infliximab cumulative dose, or infliximab effectiveness for IBD [6]. Furthermore, there was no correlation with smoking, except for palmoplantar pustulosis, which occurred more frequently in smokers (Table 2) [6]. Patients with skin lesions were more often positive for antinuclear antibodies (ANAs) [6]. In a French cohort of anti-TNF-treated IBD patients, dermatological complications were associated with a younger age at the time of initiation of anti-TNF agents, and also with Crohn's disease [10]. Pediatric anti-TNF-treated IBD patients with skin lesions displayed significantly lower fecal calprotectin levels [13]. Tillack et al. identified smoking (active or past), increased body mass index, and short IBD disease duration, but not ANAs, as predictors for anti-TNF-induced psoriasis [9]. In the study by Pugliese et al., smoking was also a risk factor for anti-TNF-induced psoriasis, whereas concomitant thiopurine use decreased the risk [19]. Smoking, female sex, and Crohn's disease were associated with anti-TNF-induced psoriasis in the study by Guerra et al. [21], while Darrigade et al. reported more severe psoriasiform eruptions in IBD patients (compared with rheumatic disease or psoriasis) and females [58]. Like Cleynen et al., Protic et al. did not detect a correlation between infliximab trough levels or anti-drug antibodies and infliximab-related psoriasis [22]. However, Huang et al. described a nonsignificant trend towards higher

Table 1 Cutaneous side effects in cohort studies with anti-TNF-treated patients

References, year	No. of patients	Disease type (n)	Patients with skin involvement (%)	Type of skin disease (n)	Type of drug (n)	Treatment discontinuation because of skin adverse events (%)
All cutaneous adverse events						
Cleynen et al., 2016 [6]	917	CD (690) UC (214) IBD unclassified (13)	264/917 (28.8)	Psoriasisiform eczema (81) Eczema (62) Xerosis (28) Palmoplantar pustulosis (14) Psoriasis (10) Infections (33) New-onset alopecia areata (5) Psoriasisiform lesions (59) Eczema (16)	Infliximab (539) Switch to other anti-TNF therapy (378)	28/917 (3.1) 28/264 (10.6)
Fréling et al., 2015 [10]	583	CD (480) UC (103)	119/583 (20.4)	Cutaneous infections (68) Psoriasis, including palmoplantar pustulosis (21) Infusion reaction + skin involvement (25) Cutaneous infections (21) Eczema (12) Xerosis (9)	Infliximab (529) Adalimumab (54)	11/583 (1.9) 11/119 (9.2)
Tillack et al., 2014 [9]	434	CD (331) UC (103)	99/434 (22.8)	Psoriasisiform lesions (68) Eczema (12) Xerosis (9)	Infliximab (387) Adalimumab (130)	7/434 (1.6) 7/21 psoriasis (33.3)
Flendrie et al., 2005 [7]	289	RA	72/289 (24.9)	Skin infections (33) Eczema (20) Ulcers (9) Psoriasis (3) Psoriasis (8) Eczema-like eruption (8) Infections (13)	TNF inhibitors [no differentiation between different TNF inhibitors]	19/289 (6.6) 19/72 (26.3)
Lee et al., 2007 [8]	150	RA (20) AS (13) PsA (1) NC (1)	35/150 (23.3)	Psoriasis (3) Psoriasis (8) Eczema-like eruption (8) Infections (13)	Etanercept (6) Infliximab (17) Adalimumab (12)	0
Mätkönen et al., 2014 [13]	84	Pediatric patients CD (64) UC (16) NC (4)	40/84 (47.6)	Psoriasisiform rash (30) Eczema (10)	Infliximab (84)	7/84 (8.3) 7/40 (17.5)
Huang et al., 2015 [12]	71	CD (46) UC (25)	18/71 (25.4)	Infusion reactions (9) Psoriasis (2) Persistent nonpsoriatic skin reactions (7) Eczema (9) Acne-like dermatitis (9) Psoriasis-like lesions (6)	Infliximab (71)	–
Baumgart et al., 2011 [14]	50	CD (46) UC (4)	31/50 (62)	Psoriasisiform lesions (6)	Adalimumab (50)	11/50 (22) 11/31 (35.5)

Table 1 continued

References, year	No. of patients	Disease type (n)	Patients with skin involvement (%)	Type of skin disease (n)	Type of drug (n)	Treatment discontinuation because of skin adverse events (%)
Włodarczyk et al., 2014 [15]	30	CD (30)	18/30 (60)	Psoriasisiform lesions (8) Eczema-like skin lesions (4) Skin dryness (4)	Infliximab (12/18) Adalimumab (4/7) Certolizumab (2/5)	–
Non-infectious skin events only						
Hellström et al., 2016 [11]	118	CD (54) UC (59) NC (5)	27/118 (22.9)	Eczema (13) Psoriasis (5) Skin dryness (2)	Infliximab (118)	0
Psoriasis only						
Guerra et al., 2016 [21]	7415	CD (5488) UC (1822) NC (105)	125/7415 (1.7)	Psoriasis (125)	Infliximab (5725) Adalimumab (3455) Both infliximab and adalimumab (1787)	45/7415 (0.6) 45/125 (36.0)
Alfali et al., 2014 [18]	1004	CD (748) UC (256)	27/1004 (2.7)	Psoriasisiform lesions (27)	Infliximab (8/620) Adalimumab (10/243) Certolizumab (9/141)	11/1004 (1.1) 11/27 (40.7)
George et al., 2015 [20]	521	IBD (521)	18/521 (3.5)	Psoriasisiform lesions (18: 16 CD and 2 UC)	Infliximab (10) Adalimumab (5) Certolizumab (3)	9/521 (1.7) 9/18 (50)
Pugliese et al., 2015 [19]	402	CD (240) UC (162)	42/402 (10.4)	Psoriasis (42)	Infliximab (242) Adalimumab (160)	11/402 (2.7) 11/42 (26.2)
Protic et al., 2016 [22]	269	CD (21) UC (2)	23/269 (8.6)	Drug-induced psoriasis (23)	Infliximab (20) Adalimumab (2) Certolizumab (1)	2/269 (0.7) 2/23 (8.7)
Dequidt et al., 2017 [59]	41	HS (41)	8/41 (19.5)	Psoriasisiform reaction (8)	Adalimumab (5) Infliximab (3)	2/41 (4.9) 2/8 (25)
Eczema only						
Esmailzadeh et al., 2009 [24]	92	CD (51) RA (21) AS (9) UC (8) PsA (3)	15/92 (16.3)	Eczema-like eruption (15)	Infliximab (92)	0

AS ankylosing spondylitis, CD Crohn's disease, HS hidradenitis suppurativa, IBD inflammatory bowel disease, NC nonclassified, PsA psoriatic arthritis, RA rheumatoid arthritis, TNF tumor necrosis factor, UC ulcerative colitis

Table 2 Risk factors for the development of cutaneous adverse events by TNF antagonists

Risk factor	Associated with anti-TNF-induced	Disease	References
Crohn's disease (vs. UC)	Cutaneous adverse events	IBD	[10]
	Non-infectious skin events	IBD	[11]
	Cutaneous adverse events	Pediatric IBD	[13]
	Psoriasis	IBD	[21]
IBD (vs. rheumatic disease or psoriasis)	Psoriasiform eruptions	IBD, rheumatic disease, psoriasis	[58]
Smoking	Psoriasis	IBD	[9, 19, 21]
	Palmoplantar pustulosis	IBD	[6]
Female sex	Psoriasis	IBD	[21]
	Cutaneous adverse events	IBD	[12]
	Cutaneous adverse events	Rheumatic disease	[25]
	Psoriasiform eruptions	IBD, rheumatic disease, psoriasis	[58]
Young age at start of anti-TNF	Cutaneous adverse events	IBD	[10]
Increased body mass index	Psoriasis	IBD	[9]
Short IBD disease duration	Psoriasis	IBD	[9]
Upper GI tract disease	Psoriasis	IBD	[20]
Perianal disease	Psoriasis	IBD	[22]
Long duration of anti-TNF	Cutaneous adverse events	Pediatric IBD	[13]
No concomitant thiopurine	Psoriasis	IBD	[19]
High dosing of anti-TNF	Skin infections	IBD	[10]
Low infliximab trough level	Infusion reaction	IBD	[12]
Adalimumab (vs. etanercept/infliximab)	Psoriasis	RA	[23]
Infliximab (vs. etanercept)	Cutaneous adverse events	Rheumatic disease	[25]
Lower fecal calprotectin level	Cutaneous adverse events	Pediatric IBD	[13]
Positive ANA	Cutaneous adverse events	IBD	[6]
Personal history of atopic disease	Eczema	IBD, RA, AS, PsA	[24]
IL-23R gene variant	Severe psoriasis or related alopecia	IBD	[9]

ANA antinuclear antibodies, AS ankylosing spondylitis, GI gastrointestinal, IBD inflammatory bowel disease, IL-23R: interleukin-23 receptor, PsA psoriatic arthritis, RA rheumatoid arthritis, TNF tumor necrosis factor, UC ulcerative colitis

infliximab trough levels in patients experiencing dermatological adverse events [12]. Low infliximab trough levels were associated with infusion reactions [12]. With respect to the TNF antagonist, there is a trend towards a lower incidence of psoriasis [23] or cutaneous side effects [25] with etanercept, compared with the TNF inhibitors adalimumab and infliximab. Similarly, Fréling et al. demonstrated that a higher dose of an anti-TNF agent was associated with a higher risk of developing cutaneous infections [10].

2.3 Classification and Skin Lesion Types

We have tried to classify anti-TNF-related skin signs in order to better understand and manage them. We identified the following categories, based on their prevalence in the Leuven cohort of IBD patients receiving anti-TNF therapy (Table 3) [6]: xerosis (41.9% of patients with skin symptoms, 10.6% of whom xerosis was the predominant sign), eczema (23.5%), psoriasiform dermatitis (30.6%), psoriasis

(3.8%), palmoplantar pustulosis (5.3%), infections (12.5%), skin cancer, alopecia and others (11%).

2.3.1 Xerosis

Patients who are taking anti-TNF agents for a longer period of time (months to years) may complain about dryness of their skin (Fig. 1), especially those who are already prone to xerosis by atopic background, older age, or during the winter season. Dry skin was observed in 41.9% of our patients with skin symptoms, 10.6% of whom had xerosis as the predominant skin sign [6]. In other patients, xerosis was accompanied by eczema or psoriasiform eczema [6], indicating that cutaneous dryness is an important instigator for the development of eczema.

2.3.2 Eczema

In our cohort, eczema or dermatitis was seen in 23.5% of IBD patients with skin signs receiving infliximab [6].

Table 3 Categories of cutaneous adverse events in patients treated with TNF inhibitors [6] (percentage of the total number of patients with skin lesions in the Leuven cohort of IBD patients receiving anti-TNF)

Cutaneous adverse events	Percentage of patients
Xerosis	41.9; 10.6 in whom xerosis was the predominant sign
Eczema	23.5
Psoriasiform eczema	30.6
Psoriasis	3.8
Palmoplantar pustulosis	5.3
Cutaneous infections	12.5
Bacterial infections: folliculitis, furuncle, cellulitis	6.4
Viral infections:	
VZV infections	1.9
Opportunistic infections: mycobacterial, fungal	very rare
Skin cancer	
Nonmelanoma skin cancer	0.4
Melanoma	0.4
Alopecia areata	1.9
Others	11

IBD inflammatory bowel disease, TNF tumor necrosis factor, VZV varicella zoster virus



Fig. 1 Xerosis cutis. Cracked appearance of the skin evolving into psoriasiform eczema with typical orange-red color

Eczema is characterized by ill-defined itchy macules, papules, or squamous lesions (Fig. 2). Bacterial superinfection is a common feature [6, 8] of eczema or psoriasiform eczema (see also below). Flendrie et al. recorded stasis dermatitis related to chronic venous insufficiency in some anti-TNF-treated RA patients [7]. Additionally, we also observed aggravated stasis dermatitis or contact dermatitis in patients treated with anti-TNF agents (personal unpublished data). Esmailzadeh et al. reported eczema in 16.3% of infliximab-treated arthritis and IBD patients, and also identified a personal history of atopy as a predictive factor for the occurrence of eczema [24]. This indicates that general measures for hydration of the skin and

avoiding skin dryness are important, especially for atopic patients receiving anti-TNF agents.

2.3.3 Psoriasiform Eczema

Psoriasiform eczema or psoriasiform dermatitis was the most frequently encountered skin sign (30.6%) in the Leuven infliximab IBD cohort [6]. It shares the characteristics of eczema (ill-defined, itchy, macular, papular, or squamous lesions), but, in addition, it displays the characteristic orange-red color reminiscent of psoriasis (Fig. 3). Histological examination shows eczema, hence the name psoriasiform dermatitis: dermatitis or eczema that looks like psoriasis, but is not psoriasis. Psoriasiform dermatitis often affects the flexural regions (retroauricular, axillary, genitocrural, submammary) mimicking inverse psoriasis, but can also be seen on the face, scalp (seborrheic dermatitis or sebopsoriasis-like), trunk, and limbs [6]. Nail lesions are rare (3.3%), adding to the fact that these lesions are not psoriasis. By contrast, bacterial superinfection very frequently leaves oozing or crusted lesions [6]. The susceptibility of skin lesions to bacterial superinfection is a striking clinical feature in these patients that is often underrated or missed in literature. It was documented in 18% of skin lesion patients in our cohort [6]; however, this is an underestimation as a bacterial swab was not systematically taken in every patient who was referred to us (personal unpublished data).

Erythematous squamous skin lesions elicited by anti-TNF are often reported as paradoxical psoriasis in the literature; however, in our experience, the majority of these patients suffer from eczema resembling psoriasis rather than psoriasis. A



Fig. 2 Eczema. Itchy ill-defined squamous lesions of the retroauricular area with oozing due to superinfection with *Staphylococcus aureus*

preponderance of eczema in inflammatory skin reactions to anti-TNF is also seen in other studies [7, 11, 14, 24]. It can be challenging to distinguish psoriasiform dermatitis from psoriasis, but, nevertheless, the lack of sharp demarcation and typical silvery-white scales, the frequent bacterial superinfection, and the rapid response to topical corticosteroids [26] help distinguish it from true psoriasis, which is definitely rarer in these patients. Histopathological examination can be performed in cases where there is doubt [6].

Adding to the confusion, some patients show lesions of both histology-confirmed psoriasiform dermatitis and psoriasis at the same time [13]. Psoriasis patients being treated with anti-TNFs may display coexisting psoriasis and psoriasiform dermatitis (Fig. 4). Guttate, and even pustular psoriasis, can sometimes be seen.

Taken together, eczema (both classical eczema and psoriasiform dermatitis) is very frequent in anti-TNF-treated patients (54% of patients with skin adverse events or 15.6% of all treated patients in the Leuven infliximab IBD cohort) [6].

2.3.4 Psoriasis

In contrast to the frequently observed psoriasiform dermatitis (30.6%), psoriasis was rare (3.8%) in our IBD patients with skin lesions receiving anti-TNF therapy [6]. Plaque psoriasis does not only have the orange-red or deep red color as in psoriasiform dermatitis but it also exhibits sharp lesional borders and silvery-white scales (Fig. 5). Superinfection is absent and itch may or may not be present. Psoriasis is more resistant to topical treatment than (psoriasiform) eczema. In some patients, psoriasis plaques are present, along with psoriasiform dermatitis [6, 13]. Guttate and pustular psoriasis may also be seen.

2.3.5 Palmoplantar Pustulosis

Palmoplantar pustulosis is characterized by recurrent crops of sterile pustules on the palms and soles (especially the nonweight-bearing areas) that evolve into erythematous squamous plaques (Fig. 6). This is one of the first described skin adverse events of anti-TNF agents [57], accounting for 5.3% of skin lesion patients in the Leuven cohort [6]. Sporadic cases of palmoplantar pustulosis (i.e. non-anti-TNF-induced) are highly linked to female sex and smoking, and,

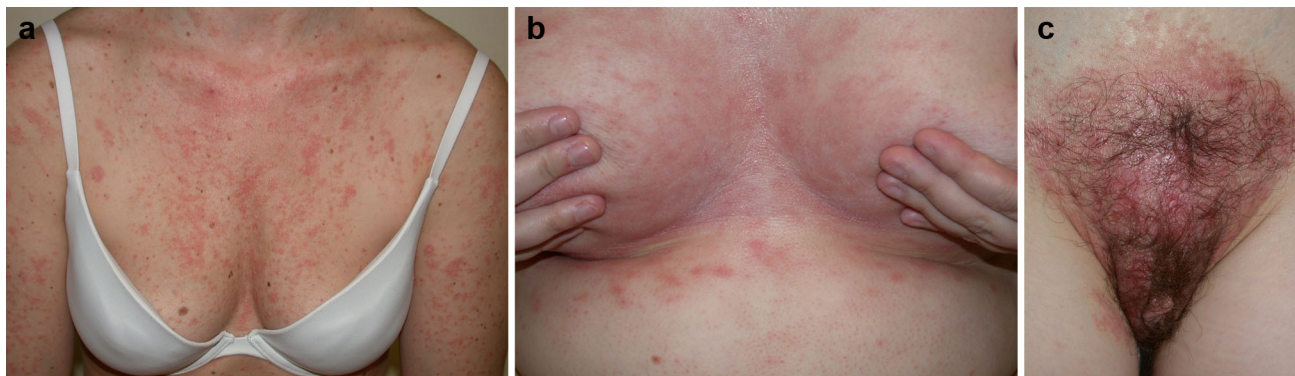


Fig. 3 Psoriasiform eczema. Ill-defined itchy macules, papules, or plaques with typical orange-red (psoriasiform) color affecting the (a) trunk and arms, (b) intermammary/submammary region, and (c) genital area, where oozing erosions indicate *Staphylococcus aureus* superinfection



Fig. 4 Psoriasiform eczema in a patient with chronic plaque psoriasis. A patient with long-standing chronic plaque psoriasis was treated with adalimumab and after some months he developed a generalized psoriasiform eczema that was confused with psoriasis. The plaque on the elbow (see *bottom right panel*) is (histologically confirmed) psoriasis with sharp borders and typical *silvery–white* scales. The generalized rash (see *top right panel*) has the *orange–red* color of psoriasis (psoriasiform) but lacks the sharp borders or typical scaling. Moreover histologic examination showed eczema, which responded rapidly to therapy with topical moderate-strength corticosteroids

although belonging to the psoriasis spectrum, palmoplantar pustulosis is nowadays considered as an entity that is clinically and genetically distinct from psoriasis [27]. Palmoplantar pustulosis induced by anti-TNF is also related to smoking, but to a lesser extent [6]; it can be accompanied by chronic plaque type, guttate, or pustular psoriasis on the rest of the body [16], and by psoriasiform eczema, the most common cutaneous finding in anti-TNF-treated patients [6].

2.3.6 Cutaneous Infections

In the Spanish registry of biologics for rheumatic disorders, infections were by far the most frequently encountered cutaneous side effects of anti-TNF agents, especially bacterial, herpes zoster, and fungal infections [25]. In our

patient cohort, cutaneous infections accounted for 12.5% of patients with cutaneous adverse events [6]. In particular, bacterial infections (folliculitis, impetigo, furuncle, and cellulitis) were not uncommon (Fig. 7) [6]. Other cohort studies report skin infections at even higher rates of up to more than half of the patients experiencing skin lesions receiving anti-TNF therapy, i.e. 21.2% [9], 37.1% [8], 45.8% [7], 48.1% [11], and 57.1% [10]. Fréling et al. described the highest percentage of cutaneous infections (57.1% of patients with skin adverse events), of which 60% were bacterial (mainly folliculitis, furuncle, nasal infection, erysipelas), 25% were fungal [mainly *Malassezia* and dermatophytes (Fig. 8)], and 13% were viral (mainly human papillomavirus [HPV], varicella zoster virus [VZV; Fig. 9], and herpes simplex virus [HSV] infections) [10]. Opportunistic infections caused by nontuberculosis mycobacteria (Fig. 10) or fungi (Fig. 11) should always be kept in mind in view of the role of TNF in the defense against intracellular microorganisms [3].

2.3.7 Skin Cancer

Ultraviolet B (UVB)-induced TNF expression by keratinocytes, resulting in apoptosis of DNA-damaged cells, is one of the protective mechanisms against photocarcinogenesis [28]; therefore, TNF inhibition may increase the risk for skin cancer. Some [25, 29], but not all [30], studies have indicated a moderately increased risk for non-melanoma skin cancer or melanoma in patients receiving anti-TNF agents [31]. Compared with rheumatic patients, IBD patients may bear a higher risk because of previous or concomitant exposure to thiopurines [32]. This may also apply to psoriasis patients due to chronic sun exposure and phototherapy [33]. Therefore, it appears cautious that patients do not undergo phototherapy while receiving anti-TNF agents, and protect themselves against UV rays with clothes or sunblocking creams. In addition, some melanoma patients experienced explosive metastasis following the initiation of an anti-TNF agent [34], indicating that TNF inhibitors are to be used with great caution, or

Fig. 5 Psoriasis. **a** Sharply defined erythematous plaque surrounded by guttate psoriasis lesions and **b** typical plaques on the scalp covered by *thick silvery–white* scales

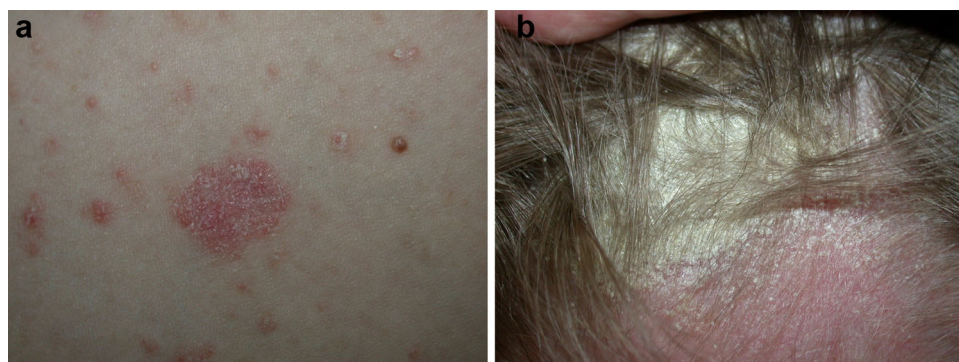


Fig. 6 Palmoplantar pustulosis. Numerous pustules affecting the (a) palms and (b) soles in repeated crops and evolving into erythematous lesions

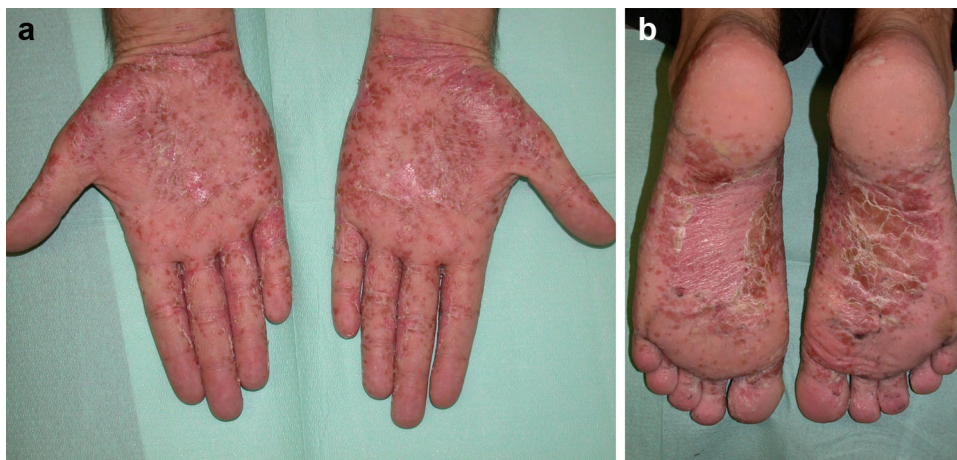


Fig. 7 Bacterial skin infections. **a** Bacterial folliculitis caused by *Staphylococcus aureus* accompanied by axillary psoriasiform eczema and **b** impetigo of the nose, again caused by *Staphylococcus aureus*



avoided if possible, in patients with a history of malignant melanoma.

2.3.8 Alopecia

Two types of alopecia can be seen in patients treated with anti-TNFs: on the one hand, alopecia secondary to psoriasis [9, 36] or psoriasiform dermatitis and alopecia areata on the other hand [6, 35].

In our experience, highly inflammatory (often superinfected) psoriasiform dermatitis of the scalp is not uncommonly accompanied by alopecia. It usually responds well to treatment of the underlying skin inflammation, although residual scarring has been reported [9, 36].

By contrast, alopecia areata occurs in the absence of visible skin inflammation. Patchy alopecia areata, but also alopecia areata totalis and alopecia areata

universalis (Fig. 12), can be encountered in patients treated with TNF inhibitors [6, 8, 35]. New-onset alopecia areata, as well as exacerbation of known alopecia areata, have been reported [6, 8, 35]. At our institution, 5 of 917 IBD patients (0.6%) initiated on infliximab developed alopecia areata [6]. In one patient, concomitant nail changes (trachyonychia) were seen [6]. In this context, a personal or family history of alopecia areata can be inquired about when considering the start of anti-TNF treatment in order to communicate a possible risk to the patient.

2.3.9 Others

Injection site reactions and infusion reactions are not uncommon in patients receiving anti-TNF agents; however, they have been discussed elsewhere [37, 38].



Fig. 8 Fungal skin infections. The genitocrural lesions in this patient could have been mistaken for flexural psoriasiform eczema but the active border raised the diagnosis of tinea cruris, which was confirmed by microscopy and fungal culture

Despite the high number of patients displaying positive ANAs while receiving anti-TNF treatment, cutaneous lupus is very rare (i.e. only 0.75% of patients with skin reactions at our

institution [6]). Lichen planus, vasculitis, and granulomatous reactions such as granuloma annulare are also rare [37, 54].

In the Leuven cohort, 3% of patients with skin reactions presented with acne, with rosacea presenting even less frequently [6]. However, Baumgart et al. reported acne-like dermatitis in 18% of IBD patients exposed to adalimumab (29% of skin reactions) [14]. One patient with hidradenitis suppurativa while being treated with an anti-TNF agent for Crohn's disease had a total remission of his skin lesions upon switching to vedolizumab, raising hidradenitis suppurativa as a possible paradoxical reaction to anti-TNF agents (personal unpublished data).

Marzano et al. recently identified amicrobial pustulosis of the fold as a cutaneous adverse event of anti-TNF agents [55, 56]. Cutaneous lymphoma developed in approximately 1 in 1000 rheumatic patients receiving anti-TNF agents [25]. Additionally, two of our patients receiving an anti-TNF agent for psoriasis developed cutaneous T cell lymphoma (personal unpublished data).

As observed by Flendrie et al. [7], we noticed compromised wound healing of a chronic venous ulcer in a patient treated with etanercept for RA. The ulcer rapidly healed



Fig. 9 Viral skin infections: recurrent varicella. This inflammatory bowel disease patient taking infliximab presented with a generalized vesicular eruption with a 'dewdrop on a rose petal' appearance, fever,

and malaise. The patient had already been through chickenpox as a child and tested positive for varicella zoster virus immunoglobulin G antibodies



Fig. 10 Opportunistic infection (mycobacterial). This inflammatory bowel disease patient on infliximab presented with an oozing and vegetating lesion on the right thumb, and nodules that slowly spread over his forearm and upper arm during the next months. The patient had an aquarium, and culture of a biopsy specimen was positive for *Mycobacterium marinum*. Accordingly, a diagnosis of fish-tank granuloma was made

when etanercept was stopped over a 4-week period for a surgical intervention, but quickly relapsed following the restart of etanercept (personal unpublished data). Ulcers accounted for 12% of skin adverse events in RA patients receiving anti-TNF treatment [7], which indicates that anti-TNF agents should be used with caution in patients with chronic venous insufficiency because of their susceptibility to venous ulcers and stasis dermatitis [7].

3 Pathophysiology

Mice lacking ADAM17 in keratinocytes display a disruption of epidermal barrier function after birth, with increased transepidermal water loss, inflammatory cell infiltration, and chronic dermatitis in adult life [39]. This has been ascribed to decreased EGFR function as ADAM17 cleaves EGFR ligands from the cell surface and topical treatment with transforming growth factor (TGF)- α reduced skin inflammation in these transgenic mice [39]. In addition, cutaneous side effects of EGFR inhibitors include



Fig. 11 Opportunistic infection (fungal). This pigeon holder was treated with adalimumab for rheumatoid arthritis and developed a wound on the upper lip evolving into an ulcerated nodule. *Cryptococcus neoformans* was identified following tissue culture

dry skin and eczema [2]. Interestingly, ADAM17 is also known as a TNF α -converting enzyme (TACE), releasing TNF from the cell membrane and pointing out a possible link between reduced TNF function, xerosis, and dermatitis.

Anti-TNF-treated patients with (psoriasiform) eczema often suffer from bacterial superinfection of their skin lesions [6]. This is also a typical feature of atopic dermatitis, whereas it is only rarely seen in psoriasis patients. Indeed, atopic dermatitis skin exhibits a decreased barrier function and decreased antimicrobial peptides, in contrast to their high abundance in psoriasis [40]. Whether bacterial superinfection of the skin of patients receiving anti-TNF agents is merely the consequence of disturbed epidermal barrier function (EGFR inhibitors also frequently cause xerosis and bacterial superinfection of the skin [2]) or changes in the innate immune system such as the reduced presence of antimicrobial peptides (such as in atopic dermatitis [40]), remains to be investigated.

A number of studies have indicated a pivotal role for interferon in anti-TNF-induced psoriasiform dermatitis [41, 42]. Indeed, the inflammatory skin changes in patients receiving TNF inhibitors very much resemble the dermatitis elicited by interferon in patients with hepatitis C [43] or multiple sclerosis (Fig. 13). Moreover, transgenic mice overexpressing interferon in epidermal keratinocytes,

Fig. 12 Alopecia areata universalis. This young adult with inflammatory bowel disease receiving adalimumab first developed a widespread psoriasiform dermatitis (*left panel*) and subsequently an alopecia areata universalis (*right panel*), with loss of not only scalp hair but also body hair, eyebrows, and eyelashes



originally designed as an animal model for psoriasis, appeared to suffer from eczema, not psoriasis [44]. Furthermore, mice overexpressing STAT3C, a transcription factor downstream of the interferon signaling pathway, also display skin changes that are strikingly similar to those seen in patients receiving TNF inhibition [45]. Likewise, imiquimod can induce or exacerbate psoriasis through toll-like receptor-7 driven production of interferon- α [46]. Tillack et al. demonstrated that it is not only the T helper (Th) 1 cytokine interferon- γ that is involved in the pathophysiology of psoriasiform dermatitis by TNF inhibitors but also interleukin (IL)-17A and IL-22 derived from Th17 cells. They conclude that the IL-12/IL-23 inhibitor ustekinumab is a valuable candidate to treat these side effects on the skin and they demonstrated this in some patients [9]. In addition, Friedrich et al. showed that keratinocyte-derived IL-36 γ and IL-17C may act to generate a pro-inflammatory self-amplifying loop in anti-TNF-induced psoriasiform skin lesions [60].

Cleynen et al. investigated known psoriasis and atopic dermatitis risk loci and found only a modest or no genetic predisposition for developing skin lesions while treated with infliximab [6]. However, Tillack et al. demonstrated that all patients with severe psoriasis or alopecia were carriers of a variant of the IL-23R gene being associated with higher Th17 cytokine production [9].

There appears to be a trend towards a lower incidence of cutaneous side effects with etanercept compared with adalimumab or infliximab [23, 25]. This may be explained



Fig. 13 Psoriasiform eczema induced by interferon. Widespread, itchy, ill-defined, orange-red macules and plaques in a patient receiving interferon injections for multiple sclerosis display a remarkable similarity to the psoriasiform dermatitis seen in anti-tumor necrosis factor-treated patients

by etanercept's properties as a soluble TNF receptor, resulting in a competitive inhibition of TNF compared with more powerful inhibition with TNF antibodies interfering more with its physiological function [3]. Accordingly, it has clearly been demonstrated that the risk for tuberculosis reactivation or opportunistic infections is higher with adalimumab/infliximab compared with etanercept [3, 47].

Table 4 Treatment scheme for cutaneous adverse events of TNF antagonists [6]

General measures and dry skin

Sun protection: use sunblocking cream, no sunbathing or phototherapy

Avoid soap, bathfoam and shower gel, but use shower/bath oil instead

Tepid water instead of hot water

Abundant use of emollients for the skin

(Psoriasiform) eczema

Avoid superinfection by decontamination of nose or anus with a disinfectant

Treat superinfection with an oral anti-*Staphylococcus aureus* antibiotic (flucloxacillin or cefuroxime)

Treat dermatitis with a topical corticosteroid (fluticasone propionate or mometasone furoate) or an antibiotic/corticosteroid combination product for superinfected eczema (betamethasone valerate + fusidic acid)

Add menthol cream and oral antihistamines (loratadine daytime, hydroxyzine night-time) in case of moderate to severe itch

Psoriasis

Treat with a topical combination product containing a vitamin D analogue (calcipotriol) and a potent corticosteroid (betamethasone dipropionate)

Acitretin, methotrexate, or cyclosporine can be used for resistant cases

Palmoplantar pustulosis

Stop smoking

Treat superinfection if present

Ultrapotent topical corticosteroid (clobetasol propionate), sometimes with the addition of salicylic acid or coal tar

Oral minocycline ($\pm$ colchicine; not in IBD)

Oral acitretin (stop minocycline)

Oral methotrexate, cyclosporine, or corticosteroids in recalcitrant cases

Discontinuation or switch of anti-TNF agent

With dedicated dermatologic care, discontinuation of anti-TNF agent is only rarely needed in patients with skin reactions (10%)

Switching between anti-TNF agents is not very useful

Switching to ustekinumab or vedolizumab may be beneficial in resistant cases

IBD inflammatory bowel disease, TNF tumor necrosis factor

4 Management

As randomized controlled trials are lacking, no evidence-based recommendations can be made for the treatment of cutaneous side effects of anti-TNF drugs. Instead, we need to rely on anecdotal reports, case series, or expert opinions (Table 4) [6, 9, 10, 21, 26].

Rahier et al. reported that more than one in three patients experiencing cutaneous adverse events discontinued their TNF inhibitor because of their skin reactions [26]. Psoriasis more often leads to discontinuation of anti-TNF treatment than cutaneous infections [10]. At our institution, almost 90% of patients with dermatological skin symptoms were able to continue therapy with dedicated dermatologic care [6], which clearly underscores the importance of accurate management of these patients.

4.1 General Measures

Adequate sun-protective measures (hat, clothes, sunblocking creams) should be advised as anti-TNF treatment may

accelerate photocarcinogenesis. In addition, exposure to artificial UV needs to be avoided (sunbed, and UVB and psoralen and ultraviolet A [PUVA] therapy).

In order to prevent cutaneous dryness, the patient is instructed to avoid skincare products or habits that dry out the skin (such as bath foam, shower gel, soap, very hot water, and frequent bathing or showering) and to switch to bath/shower oil and lukewarm water. An emollient cream is to be used on a daily basis, especially following a bath/shower, swimming, or a sauna. Special attention should be given to patients already suffering from atopic dermatitis or dry skin [24].

Dermatologists are very familiar with the treatment of eczema (whether or not superinfected) and psoriasis, and referral to a dedicated dermatologist is therefore recommended [6, 10, 48]. A majority of patients can be helped with topical treatment only [6, 9–11, 48].

4.2 (Psoriasiform) Eczema

A moderately potent topical corticosteroid such as fluticasone propionate or mometasone furoate can be prescribed

for eczema in a taper down scheme (e.g. once daily for 2 weeks, three times a week for 2 weeks, twice a week for 2 weeks, and once a week for 2 weeks, with application of an emollient cream on the other days). Moreover, a topical menthol-containing cream and oral antihistamines (e.g. loratadine at daytime and hydroxyzine at night) can be used to relieve itch [6].

Tackling bacterial colonization and superinfection is pivotal in the management of patients with eczema. A fixed combination of a corticosteroid with an antibiotic can be used (e.g. betamethasone valerate plus fusidic acid cream) upon bacterial superinfection. Additionally, a course of oral antibiotics against *Staphylococcus aureus* can be initiated (e.g. flucloxacillin or cefuroxime), often resulting in a dramatic improvement of the skin lesions. In this case, eradication of *Staphylococcus aureus* carriage by long-term daily application of a chlorhexidine-based disinfectant to the nose and anus is also an important treatment goal [6].

4.3 Psoriasis

Plaque psoriasis can be treated with a fixed combination of calcipotriol and betamethasone dipropionate. In cases of treatment resistance, oral acitretin (which may aggravate skin dryness), methotrexate, or cyclosporine can be used [6].

4.4 Palmoplantar Pustulosis

Smoking cessation is recommended and an ultrapotent corticosteroid (clobetasol propionate) combined with salicylic acid or coal tar in a cream base is prescribed. Next, oral tetracyclines can be used along with colchicine (except in IBD patients). Acitretin, methotrexate, or oral corticosteroids are used for recalcitrant cases [6].

Tillack et al. described response to ustekinumab in two patients with anti-TNF-induced palmoplantar pustulosis. However, both patients received oral alitretinoin concomitantly, and one patient experienced partial recurrence following cessation of alitretinoin [9].

4.5 Alopecia

Alopecia secondary to superinfected psoriasiform dermatitis responds to treatment of the underlying eczema or psoriasis [9, 36]; patchy alopecia areata can be treated by ultrapotent topical corticosteroids. Treatment with a contact sensitizer such as diphenylprone is not recommended because of the aggravated eczematous response in anti-TNF-treated patients. In rapidly progressive alopecia areata, alopecia totalis or alopecia universalis, a switch to ustekinumab or vedolizumab should be considered. Anecdotal reports have associated ustekinumab with mild forms of

alopecia areata [49]. Conversely, ustekinumab has also been convincingly shown to be very promising for the treatment of extensive alopecia areata in a case series [50]. Vedolizumab appears to be neutral towards alopecia areata (personal unpublished data) [53].

4.6 To Switch or Not to Switch Anti-Tumor Necrosis Factor Agents

With dedicated dermatologic care, cessation or interruption of the causal anti-TNF agent due to cutaneous side effects is only rarely needed [6]. Interruption of the anti-TNF agent is only recommended when unbearable cutaneous reactions persist despite optimal dermatological care. Furthermore, as the skin signs take considerable time to arise after the start of a TNF inhibitor [6], they also take time (mostly numerous months) to subside when the drug is stopped.

Switching between different anti-TNF antibodies is not a very useful strategy in the management of cutaneous side effects [6, 9, 10]. This is not unexpected as cutaneous adverse events are a reflection of the inhibition of the physiological function of TNF in the skin, and is therefore class-specific. Being a somewhat less powerful anti-TNF drug [3], etanercept brings about skin problems to a lesser extent [23, 25]. Tillack et al. focused on the beneficial effect of switching to ustekinumab as it does not provoke xerosis, (psoriasiform) dermatitis, psoriasis or palmoplantar pustulosis, and displays therapeutic effects on both the Th17 axis and psoriasis [9]. Despite some case reports of induction of patchy alopecia areata [49], ustekinumab showed very promising regrowth in some patients with extensive alopecia areata [50], making it an interesting future candidate for this indication.

Switching to vedolizumab is another possibility [51, 52]. It does not induce skin problems but is not expected to demonstrate benefit on existing skin conditions or other extraintestinal manifestations of IBD [53]. As it exhibits a gut-specific mechanism of action, vedolizumab may be beneficial in patients with high susceptibility to cutaneous infections.

5 Conclusions

We reviewed the literature on anti-TNF-induced skin lesions and added our personal experience in providing dermatological care to these patients. Approximately one in four patients receiving anti-TNF treatment presents with cutaneous adverse events, most often dry skin, (psoriasiform) eczema, psoriasis, palmoplantar pustulosis, skin infection, alopecia, or skin cancer. The mainstay of therapy of psoriasiform eczema consists of topical corticosteroids and oral antibiotics to treat bacterial superinfection.

Discontinuation of anti-TNF therapy is only rarely required upon adequate dermatological treatment of these skin lesions.

Smoking, female sex, and Crohn's disease are most frequently observed as risk factors for anti-TNF-induced cutaneous adverse events. The underlying pathophysiology is still poorly understood but epidermal permeability barrier dysfunction, increased susceptibility to bacterial superinfection and cytokines derived from Th1 (interferon- γ), Th17 cells (IL-17A and IL-22), plasmacytic dendritic cells (interferon- α), and keratinocytes (IL-36 γ and IL-17C) appear to play a role.

Further research into the risk factors and pathophysiology of these troublesome skin lesions may help to find better strategies for prevention and/or treatment.

Author contributions Siegfried Segaert and Caroline Hermans prepared and wrote the manuscript.

Compliance with Ethical Standards

Funding No funding was received for the preparation of this review.

Conflict of interest Siegfried Segaert and Caroline Hermans have no conflicts of interest to declare.

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